

Application of Deep Learning in the Identification of Extrachromosomal Circular DNA (eccDNA)

Maolin Zeng
School of Life Sciences and
Engineering
Southwest Jiaotong University
Chengdu, Sichuan, China
zeng1716220414@my.swjtu.edu.cn

Yuguo Li
School of Life Sciences and
Engineering
Southwest Jiaotong University
Chengdu, Sichuan, China
3211239335@qq.com

Zhiyun Guo*
School of Life Sciences and
Engineering
Southwest Jiaotong University
Chengdu, Sichuan, China
zhiyunguo@swjtu.edu.cn

Abstract

EccDNA has emerged as a pivotal factor in regulating oncogene expression. Its diverse structures and rich variability present significant challenges for accurate identification. In this study, we propose a composite deep learning model that integrates convolutional neural networks (CNNs), residual networks, and attention mechanisms. Leveraging existing eccDNA datasets, the model is designed to automatically detect eccDNAs across various samples, with the goal of enhancing recognition efficiency and precision. Upon training, the model demonstrated a significant improvement in eccDNA identification. Additionally, we developed a software tool named ECCNET to facilitate access and application by researchers. Validation experiments confirm that the model exhibits outstanding performance in eccDNA sequence recognition and analysis, maintaining a high degree of consistency across different datasets. This research effectively applies advanced deep learning techniques to eccDNA identification, laying a solid foundation for further exploration in this field.

CCS Concepts

• **Computing methodologies** → Machine learning; Machine learning approaches.

Keywords

EccDNA, CNN, residual networks, attention mechanism, GAN

ACM Reference Format:

Maolin Zeng, Yuguo Li, and Zhiyun Guo. 2025. Application of Deep Learning in the Identification of Extrachromosomal Circular DNA (eccDNA). In *2025 2nd International Conference on Modeling, Natural Language Processing and Machine Learning (CMNM 2025)*, May 16–18, 2025, Fuzhou, China. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3757110.3757175>

1 INTRODUCTION

EccDNA is a type of circular DNA that may exist independently of chromosomes [1–3], with sizes ranging from hundreds of base pairs to several megabases [3, 4]. Recent studies have demonstrated that eccDNA plays a pivotal role in cancer pathogenesis [1], the

evolution of tumor heterogeneity [5], and the development of therapeutic resistance [6], thereby highlighting the importance of the precise identification and investigation of eccDNA in cytogenetics and genomics. As eccDNA research has advanced, a vast amount of sequencing data and eccDNA sequences generated by identification tools has been extensively accumulated [1, 7]. Although several existing tools [8, 9] (e.g., Circle-Map, Circle-Finder, and AmpliconSuite-pipeline) are designed to identify eccDNA based on its circular structure, these methods are often constrained by their pattern-matching algorithms or feature extraction techniques, potentially missing critical information when processing large-scale data [7, 10]. However, the high dimensionality, complexity, and heterogeneity of these data render traditional analytical methods insufficient for fully uncovering the underlying sequence features and biological significance.

The rapid advancement of deep learning has introduced new opportunities for biological research. Leveraging its ability to recognize complex patterns in large-scale data, deep learning has demonstrated exceptional performance in genomics and proteomics [11–13]. Consequently, applying advanced machine learning methods to explore the latent patterns of eccDNA has emerged as a novel research direction. Building on these advancements, we developed a software tool named ECCNET, which integrates CNN, multi-head attention mechanisms, and residual networks to predict and analyze the circular structure of eccDNA. ECCNET efficiently processes large-scale eccDNA data, automatically extracts and identifies critical features through its sophisticated network architecture, thereby improving both the accuracy and efficiency of identification. Compared to traditional pattern-matching algorithms, ECCNET not only handles highly heterogeneous biological data but also maintains high consistency across diverse datasets, demonstrating its unique advantages in eccDNA research. By using ECCNET, the identification and analysis of eccDNA have achieved significant enhancements in accuracy and efficiency, providing an essential tool for in-depth exploration of the roles of eccDNA in cancer pathogenesis, tumor heterogeneity evolution, and therapeutic resistance development.

2 METHODS

2.1 Model Construction of ECCNET

The Pyfaidx tool was used to extract eccDNA sequences from existing databases. For unsupervised learning samples, one-hot encoding was applied. For supervised learning samples, both one-hot encoding and DNA2vec encoding methods were implemented. In

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License. *CMNM 2025, Fuzhou, China*

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1434-4/2025/05

<https://doi.org/10.1145/3757110.3757175>

the case of large-scale supervised learning samples, a cyclic padding-based one-hot encoding method was employed. Specifically, the maximum length within the sample set was used as the row dimension of the encoding matrix, and sequences shorter than the maximum length were cyclically padded. This approach preserved the circular structural features by duplicating the padded encoding at both the beginning and the end of the sequence. For small-scale supervised learning samples, the DNA2vec encoding method was utilized. The eccDNA prediction model employed three variants of a one-dimensional convolutional neural network (1DCNN) architecture—Origin, Deep, and Simple—each tailored for datasets of different scales. Origin Model contains three convolutional layers with 32, 64, and 128 filters, respectively. The kernel size is set to 5, and the activation function is ReLU. Each convolutional layer is followed by a max-pooling layer (pool size = 2). A flattening layer converts the convolutional output into a one-dimensional vector, followed by a fully connected layer with 256 neurons and a dropout layer (dropout rate = 0.2). The output layer consists of a single neuron using the sigmoid activation function for binary classification. Deep Model is designed for complex eccDNA prediction and large-scale data training. This model builds upon the Origin Model by increasing the number of filters to 64, 128, and 256. The kernel size is reduced to 3. The fully connected layer contains 512 neurons, and the dropout rate is increased to 0.5. Simple Model is suitable for small-scale datasets. It includes a single convolutional layer with 32 filters (kernel size = 3), followed by a max-pooling layer (pool size = 2). The flattening layer is connected to a fully connected layer with 64 neurons and the output layer uses the sigmoid activation function for binary classification.

2.2 Machine Learning for eccDNA

The autoencoder model employ a multi-layer fully connected neural network architecture, consisting of an input layer, multiple hidden layers, and an output layer. The hidden layers use the ReLU activation function to perform nonlinear transformations. The model is compiled with the Adam optimizer, and the mean squared error (MSE) is selected as the loss function. Unless otherwise specified, the model is trained for 50 epochs. A 32-dimensional feature vector is obtained from the output of the encoding layer, representing the dimensionality-reduced form of the input data. This feature vector is used as input for clustering analysis. The K-means clustering algorithm is applied to the encoded data, with the number of clusters initially set to two; this parameter is adjustable based on specific experimental requirements. The t-distributed stochastic neighbor embedding(t-SNE) algorithm is used to perform dimensionality reduction and visualization of the encoded data.

2.3 Data Collection and Distribution Analysis

Human eccDNA data are collected from the three public databases—eccDNAAtlas, eccbase, and circlebase—totaling 821,664 records [14–16]. Duplicate eccDNA sequences are removed using the bedtools tool. The length and density distributions of eccDNA are analyzed using the ggplot2 package.

2.4 Data Augmentation

For small-sample data, a Generative Adversarial Network (GAN) is used to augment the dataset. A Deep Convolutional Generative Adversarial Network (DCGAN) architecture is employed, in which both the generator and discriminator are implemented using CNNs [17]. The generator takes as input a random noise vector (latent vector) with a dimension of 100. The first fully connected layer expands the input to 128 feature channels and uses the LeakyReLU activation function. Multiple one-dimensional transposed convolutional (Conv1DTranspose) layers, each applying the LeakyReLU activation function. The final layer uses the tanh activation function to produce the output. The discriminator receives as input a tensor with shape (1000, 4). It applies multiple one-dimensional convolutional layers, each followed by the LeakyReLU activation function. These layers are followed by a flattening layer, and the final output layer uses the sigmoid activation function to classify the input as real or generated.

2.5 Construction of a Complex 1D Convolutional Neural Network

A complex 1DCNN model is constructed for eccDNA binary classification tasks. The model integrates residual blocks and multi-head self-attention mechanisms to increase its depth and representation capacity [11, 12]. Residual blocks are incorporated using skip connections to facilitate information flow. Each residual block contains two convolutional layers with a kernel size of 3 and 64 or 128 filters. Each convolutional layer is followed by batch normalization and applies the ReLU activation function. A residual connection is formed by adding the input tensor to the output tensor after the convolution operations. A Multi-Head Self-Attention mechanism is embedded within the architecture. Each attention block consists of four heads, with each head having a key dimension of either 64 or 128. Layer normalization is applied after the multi-head attention computation.

Additionally, a customized loss function is implemented based on the standard binary cross-entropy loss, incorporating prediction confidence as a weighting factor. Specifically, the absolute difference between the prediction and 0.5 is used to compute confidence. The standard binary cross-entropy loss is then multiplied by this confidence value to adjust the loss. The standard binary cross-entropy loss function is defined as follows:

$$\mathcal{L}_{\text{bnr_cosnrp}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (1)$$

Where y_i denotes the true label of the i^{th} sample, tp_i denotes the predicted probability of the i^{th} sample, and N denotes the total number of samples. The customized loss function is defined as follows:

$$\mathcal{L}_{\text{cso}} = -\frac{1}{N} \sum_{i=1}^N \left[\left(\frac{y_i \log(p_i)}{+(1 - y_i) \log(1 - p_i)} \right) \cdot 2 |p_i - 0.5| \right] \quad (2)$$

In this formulation, the confidence weighting factor $2|p_i - 0.5|$ amplifies loss associated with predictions further from the decision boundary.

2.6 Model Training and Evaluation

Negative samples are generated by randomly selecting genomic regions with less than 20% overlap with eccDNA regions[1]. The length distribution and chromosomal distribution of negative samples are matched to those of the eccDNA samples, using the default random seed of 42. Data are loaded through a data generator with a batch size of 32 samples. After encoding, positive and negative samples are labeled as 1 and 0, respectively. Subsequently, 10% of positive and negative samples are separately selected as the validation set, 5% are used as the test set, and the remaining samples are combined to form the training set. Sample encoding strategies and model selection are controlled by setting the parameters `deep` and `encode_type`, respectively. The number of training epochs is specified using the parameter `e`. Upon completion of training, the model is saved in .h5 format. Model performance is evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. Training dynamics are visualized using line plots, overall performance is presented using bar charts, and comparative performance across different models is illustrated through radar charts.

3 RESULTS

3.1 Construction and Application of ECCNET

With increasing recognition of the critical role eccDNA plays in diseases such as cancer, how to correctly and quickly identify eccDNA becomes particularly urgent. However, the existing identification tools usually confine themselves to use pattern matching or feature extraction methods which are hard to deal with huge volumes of complicated biological data and easy to ignore some crucial information. What's more, existing eccDNA identification methods based on deep learning such as Deepcircle have a limitation under the condition of sequence length restriction.

Considering the above situation, ECCNET is a tool that combines supervised learning and unsupervised learning for predicting eccDNA. ECCNET allows users to choose between selecting any pre-constructed model as well as choosing the option of uploading user data for further modeling, reconstructing models based upon same as well as fully predicting and analyzing eccDNA (Figure 1a). ECCNET integrates CNN, residual networks, and multi-head attention mechanisms to construct training and prediction modules capable of executing binary classification tasks. ECCNET designs training and prediction models according to CNN, residual networks, multi-head attention mechanism; constructs the analysis model of sample differences according to unsupervised learning model. Therefore, combining the advantages of deep learning technology, not only can we predict the sequences of eccDNA but also find possible difference and regularity between samples by analysis mode such as unsupervised learning (Figure 1b).

High-quality datasets are the foundation of further modeling tasks carried out by deep learning models. When identifying eccDNA, it always has characteristics such as long variable length and structural diversity in nucleic acid base sequences, and at the same time, the accurate circular coding of eccDNA has great influence on the final model effect, so it is very difficult to carry out model training and feature extraction. In order to construct the maximum variation of the training set and test the degree of effectiveness of ECCNET, combined with public database researches on

821664 different eccDNAs collected from eccbase, circlebase, and eccDNAAtlas databases.

Besides, since the data ECCNET could deal with had great quantity, plus the complex composition of eccDNA and the heterogeneous diversity brought by multiple origins, we specially coded some sequences related to circular features of eccDNA to capture valuable information and more distant dependencies well (Figure 1d). The combination of supervised learning and unsupervised learning made ECCNET a convenient tool for studying eccDNA. Special codes were only used when partial feature loss needed to be compensated for, but due to the special nature and wide distribution of eccDNA, it was necessary to further build a better and deeper new structure based on the first ECCNET model in order to better grasp the specific difference contained in eccDNA (Figure 1c). At this stage, in the optimized version of the original model after adding residual blocks that improved the thickness and transmission distance between layers, solved the gradient disappearance problem during training; the multi-head attention mechanism guaranteed sufficient capture of the target area through focus, fully considering the dependence relationship of information within a certain distance; at the same time, according to the importance weight corresponding to each base pair point predicted, customized a loss function emphasizing predictions based on confidence levels, which improved the final effect close to the research method of solving mathematical problems in biology. Therefore, ECCNET was significantly different from existing tools and provided strong support for researchers' in-depth studies on eccDNA.

3.2 Genomic Distribution Analysis of eccDNA Datasets

To encode the circularity of eccDNA, two special sequence encoding methods are proposed: one-hot and DNA2vec. Namely, the length of one-hot equals to the max size of all eccDNAs and circular encoding is used. One-hot pads both ends' features of eccDNA into consideration for physical and circular structure information capturing, which is conducive to important feature information catching. DNA2vec captures complex feature information by learning hidden k-mers contained in a sequence, so as to facilitate unsupervised discovery of latent difference and population structure information between samples.

Dataset diversity can lead to more complete model training and better generalization ability. The above -mentioned specific coding modes can improve the ability of recognizing useful information by taking account of its own characteristics(circular). Specifically, compared with the traditional one-hot mode, DNA2vec exhibits significant superiority in improving the model.

The following analyses will be used to determine whether the obtained eccDNA data is reliable and what their distribution features may be. Through analysis, it has been confirmed that many sequences' lengths are located between 100 bp and 10Kb (Figure 2a); however, there also exist a few eccentric long sequences over several megabases. In addition, the density heatmap demonstrates that some chromosomes exhibit densely distributed eccDNA numbers (Figure 2b). For instance, chromosomes 17 and 19 display dense eccDNA distribution points. From this point of view, we suppose that areas where large amounts of eccDNAs distribute have certain

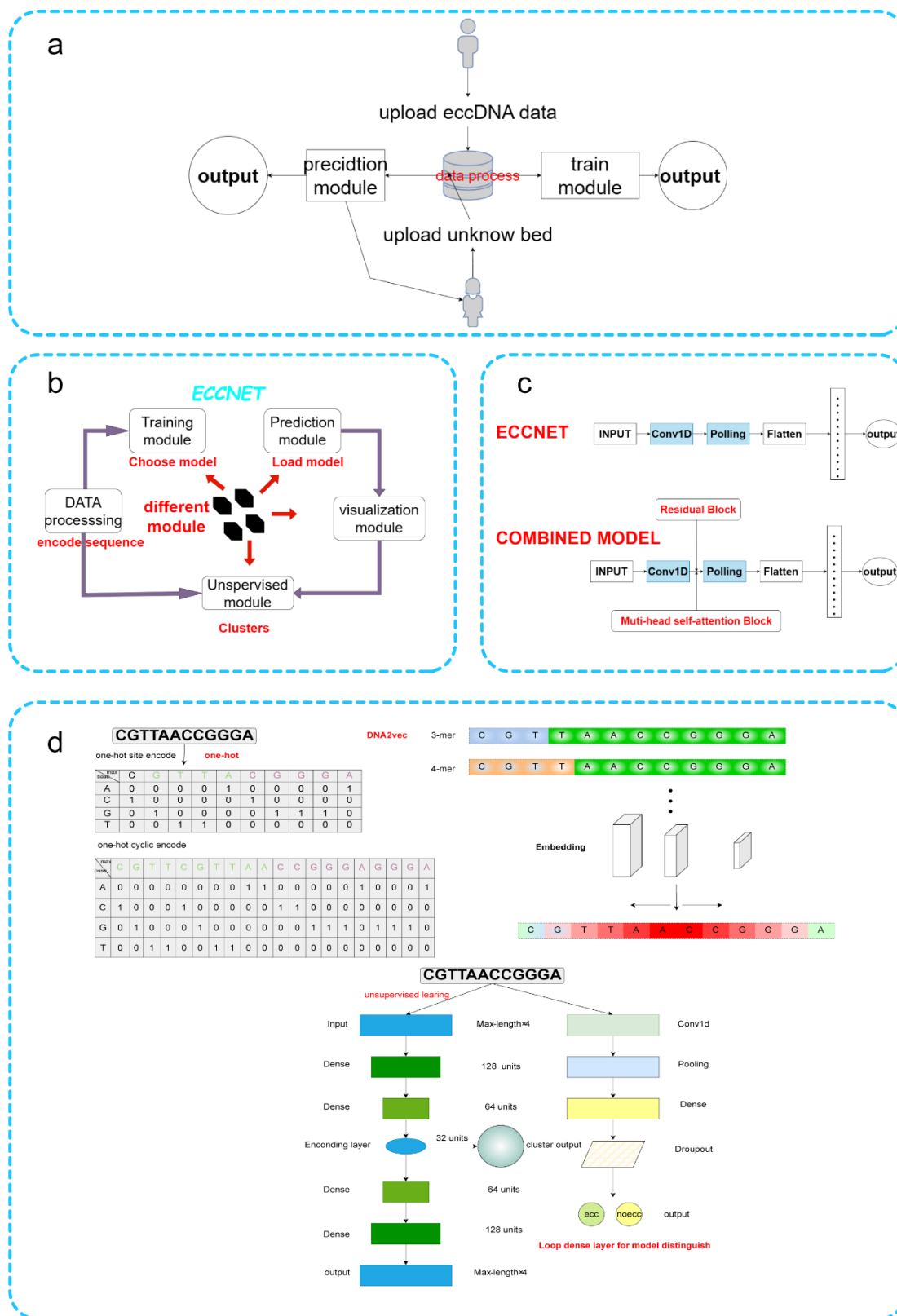


Figure 1: Construction flow chart of ECCNET and application diagram. (a) Application flowchart of ECCNET. (b) Schematic diagram of functional architecture of ECCNET. (c) Data encoding scheme illustration (d) Drawing sketch of construction flow of complex model.

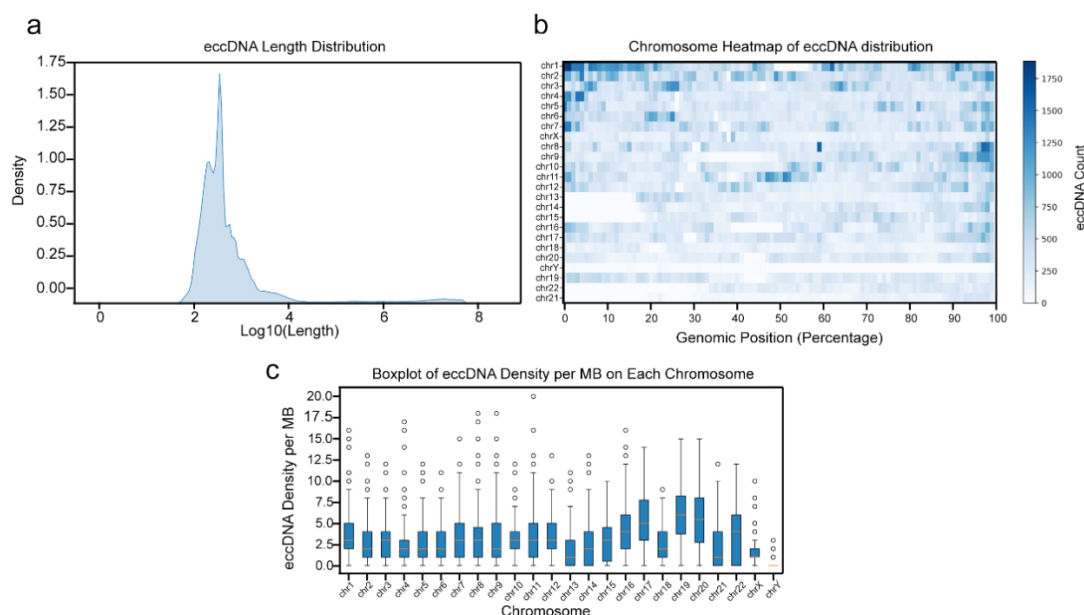


Figure 2: Distribution of eccDNA across the genome. (A) Cumulative length distribution of eccDNA dataset. (B) Chromosomal density heatmap of eccDNA dataset. Y axis is chromosome name and X axis indicates chromosomal positions using percentage of whole chromosome length; color denotes number of eccDNAs inside every region. (C) Density distribution of the quantity of eccDNA across different chromosomes. X axis denotes chromosome names and Y axis denotes the eccDNA number per MB of each chromosome.

biological meaning, which is consistent with results discovered in previous works. The cross-chromosome eccDNA quantity density map (Figure 2C) provides a more intuitive distribution of eccDNA on different chromosomes and further confirms this conclusion.

3.3 Machine Learning Analysis of eccDNA

To detect hidden sequence variability and clustering pattern among eccDNA samples, we first used unsupervised learning for the collected eccDNA dataset by autoencoder-aided feature extraction and DNA2vec-encoding dimensionality reduction. As eccDNA sequences are one-hot encoding at first place, we use autoencoders to perform dimensionality reduction for high-dimensional eccDNA data; however, as the autoencoder mostly converts different sequences into uniform lower-dimensional vectors that cannot clearly show the hidden structures of these original data, then we employ t-SNE to reduce high-dimensional eccDNA data into two-dimension data while keeping local neighborhood relationship (Figure 3a). EccDNA samples have evident distribution modes in this reduced-dimensional space showing considerably varying sequences among eccDNA samples. To further obtain potential clustering patterns and inherent correlations between eccDNA samples, here we applied K-means clustering algorithm on such reduced-dimensional eccDNA data (Figure 3b), where eccDNA samples can be divided into several relatively well-separated clusters that have quite differing sequences from each other, revealing evidence supporting effective utilization of machine learning methods in the analysis of eccDNA-related information.

For a better display of obtained clusters with analyzing the separating abilities of local samples from adjacent ones and exploring latent sequences within the entire eccDNA dataset, here we applied DNA2vec-encoding feature vectors and investigated the resulting clustered patterns of the same datasets but utilizing DNA2vec encodings (Figure 3c) as shown previously. At last, to investigate the discriminative ability between eccDNA and non-eccDNA samples, we also applied DNA2vec encoding on randomly sampled DNA fragments and highly confident eccDNA sequences; as expected, we observed distinct and separated distributions between eccDNA samples and non-eccDNA random DNA samples in our analyzed figure (Figure 3d), confirming reliable discriminations provided by DNA2vec encoding method.

Moreover, we trained supervised learning with the eccDNA dataset and calculated the improved accuracy of the ECCNET model. We used the original ECCNET to encode, train, and validate all collected eccDNA samples and adapted the base model to handle varying sample numbers. Testing on multiple datasets demonstrated that each method resulted in a small but cumulative increase in accuracy and enabled gradual loss reductions during model training (Figure 3e). ECCNET achieved greater accuracy (0.8713) and lower loss (0.1942) than other methods (Figure 3f)[7, 10]. The ROC curve showed good model performance (Figure 3g, AUC=0.96). At the same time, it is noteworthy that data encoding methods found that DNA2vec significantly increased model performance (Figure 3g). This proves that machine learning's encoding methods can distinguish between eccDNA and non-eccDNA samples. This finding highlights the effectiveness and precision of deep learning models

for classification problems. It suggests using unsupervised learning more broadly in eccDNA studies and provides useful leads for further functional research.

Although the ECCNET base model works well for eccDNA recognition problems, the complexity and high-dimension characteristics of eccDNA data pose challenges for deeper investigation using deep learning methods. Furthermore, the performance of traditional deep learning models often depends on the size of the training dataset; however, eccDNA datasets are limited in number. Therefore, exploring deeper network models and suitable data enhancement methods will be necessary for further optimizing ECCNET's performance.

3.4 Performance Improvement of ECCNET Complex Model and Data Augmentation

Then we carry out more ECCNET tests for complex function refinement and prove the advantage of the ECCNET complex model for detecting eccDNA. On the basis of the original basic ECCNET, we add residual network blocks, attention layer structures, and a custom loss function to form an improved ECCNET complex model. Because there are certain defects in the eccDNA dataset itself, such as small size and the complexity of high-dimension features, the model meets some problems in generalization ability and accuracy when dealing with real data. Therefore, we use GAN for data augmentation; consequently, the training dataset is enlarged. Through further experiments compared with traditional models (logistic regression (LR), decision tree (DT), random forest (RF), and support vector machine (SVM)) the results showed that ECCNET had better performances [accuracy, precision, recall, and F1 score] than these models (Figure 4a). Then, comparative experiments were conducted using simpler models: LR, DT, RF, and SVM. The complex ECCNET displayed higher accuracies, precision, recall, and F1-scores than other simple models (Figure 4a), demonstrating its ability to deal with high-dimensional eccDNA data. We also confirmed this by comparing ROC curves and found that our ECCNET achieved a higher AUC score than the previous method did.

Finally, we tested each module's contribution through ablation analysis. When removing residual blocks gradually or deleting attention layers, regularization, and self-defined loss functions one by one from the ECCNET complex model, there are obvious decreases in different metric values, especially after deletion of the regularized block, resulting in sharp drops in both accuracy and recall rates (Figure 4b & 4c). This confirms the importance of adding the regularized block and the significance of the residual network blocks and attention layers in achieving good performance.

Considering the limitations caused by datasets like insufficient quantity and high dimensionality, we take advantage of the benefits of GAN data augmentation. With high cost and low availability of eccDNA data, GANs generate simulated data similar to actual ones, which can be used to increase the training set size. Combined with synthetic GAN-generated data and authentic data sets, we retrained ECCNET (Figure 4d); it demonstrated better classification accuracy with much-reduced error rates, thus proving that the simulated data generated by GANs effectively enlarges the scale of the training set and improves the robustness and generalization capacity of the ECCNET application scenario.

3.5 Performance Comparison between ECCNET and Other Deep Learning Methods

In order to sum up the general situation of the performance of ECCNET in predicting eccDNA, we carried out ablation tests and comparison tests with other algorithms such as SVM and DeepCircle. The existing bioinformatics models cannot adapt to the complexity and diversity of eccDNA. Comparison testing is used to verify whether or not ECCNET can obtain better results than traditional methods in multiple indicators, which verifies the effectiveness and reliability of processing biological information by using ECCNET.

Firstly, the same dataset was used to train and test the basic model of ECCNET, and the basic model of ECCNET was tested with DeepCircle and classic models respectively. At the same time, the input sequence was encoded by two methods (one hot and DNA2vec), and at the same time, we performed ablation tests on the basis of the ECCNET model. Specifically, this process adopts a complex model for ECCNET coding and the circle padding method as the initial state value; then, eliminate one part after another to see if the accuracy decreases.

The result shows that our baseline model is better than comparative methods under every index, especially under accuracy, precision, and F1 score (Figure 5a). It also proves that after introducing the DNA2vec algorithm, its model enhancement effect has been verified. Compared with traditional features (the one-hot mode), the DNA2vec feature improves the recognition ability (Figures 5b & 5c). Similarly, in order to observe how the ablation experiment affects the final prediction, we carry out an ablation experiment based on the complex ECCNET model, specifically taking the circular padding code as the base group value and eliminating parts gradually; draw the line plot (Figures 5d & 5e).

These conclusions show that ECCNET is effective and reliable in predicting eccDNA with high-accuracy rate; it will be superior to some traditional tools, even when considering the computing cost, as well as having relatively high speed. Incorporating the DNA2vec encoding method into the model enables us to obtain a more accurate predictive effect and enhances the extractability and representation capacity of special elements in the data set through the combination of convolution and other operations. This aspect confirms the validity of ECCNET as an excellent model designed for eccentric circles research.

4 CONCLUSION

ECCNET significantly enhances eccDNA identification by integrating CNN, residual blocks, multi-head attention, DNA2vec encoding, and a bespoke loss function, resulting in improved accuracy, precision, recall, and F1-score across several public datasets. Unsupervised clustering and GAN-based augmentation reveal significant eccDNA trends and enhance performance with limited samples. Ablation studies confirm the critical roles of residual and attention components in the model. Dense eccDNA clusters localized in specific chromosomal regions are implicated in cancer pathogenesis, tumor heterogeneity, and therapeutic resistance. Future research should focus on expanding ECCNET's validation, incorporating additional biomarkers and optimizing data augmentation strategies. In conclusion, ECCNET provides a versatile and scalable framework

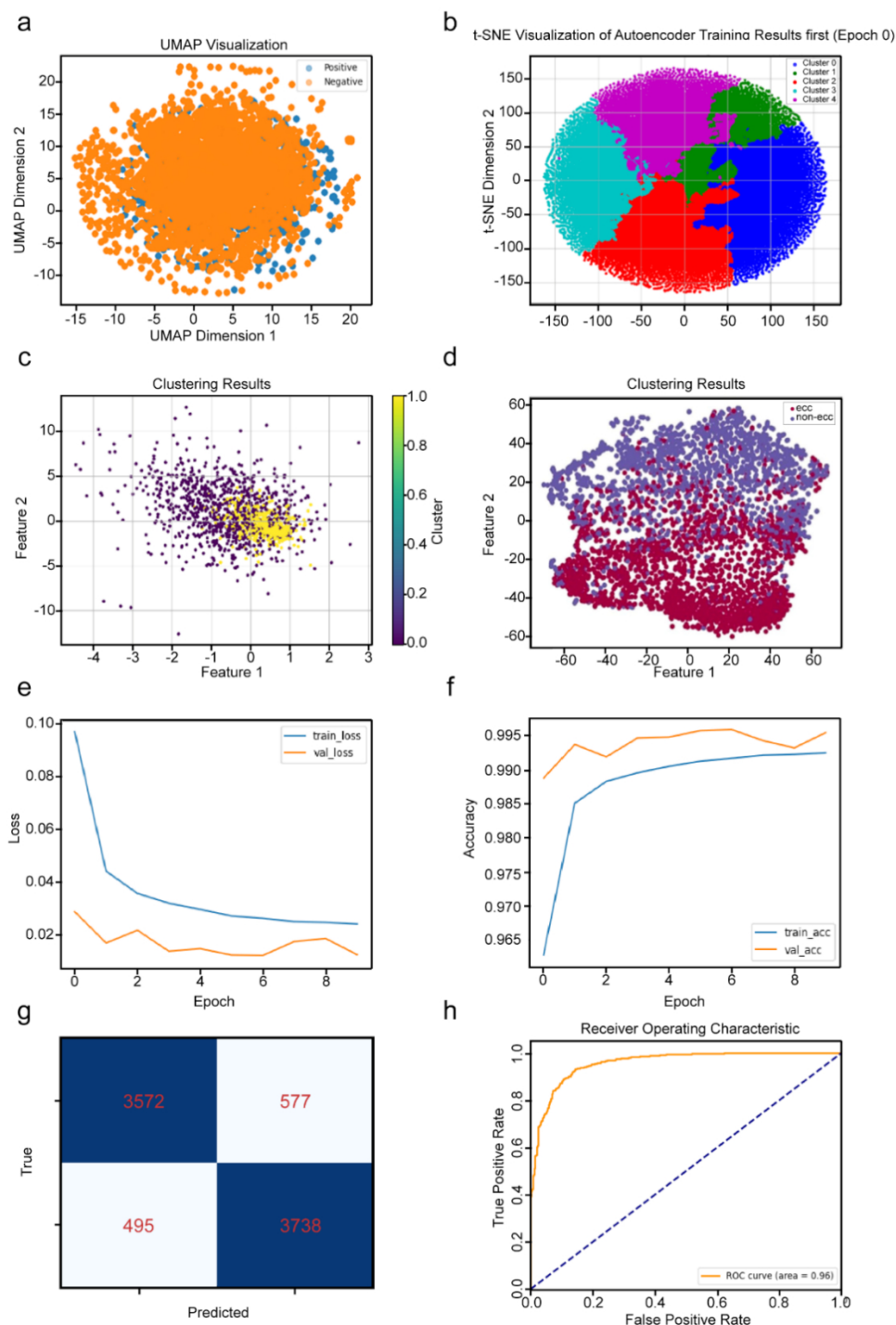


Figure 3: Clustering analysis of eccDNA after encoding. (a) UMAP clustering result of all positive and negative samples encoded by DNA2vec. (b) eccDNA clustering result based on autoencoder dimensionality reduction. (c) Two-dimensional clustering result of randomly generated samples. (d) Clustering result of eccDNA samples from human embryonic cells and randomly generated negative samples. (e) Training and validation loss curves over 10 rounds of training. The blue line represents the training loss value; the orange line represents the validation loss value. (f) Training and validation accuracy curves over 10 rounds of training. The blue line represents the training accuracy value, while the orange line represents the validation accuracy value. (g) Confusion matrix representing binary classification results on the test set. (h) ROC curve of the test set.

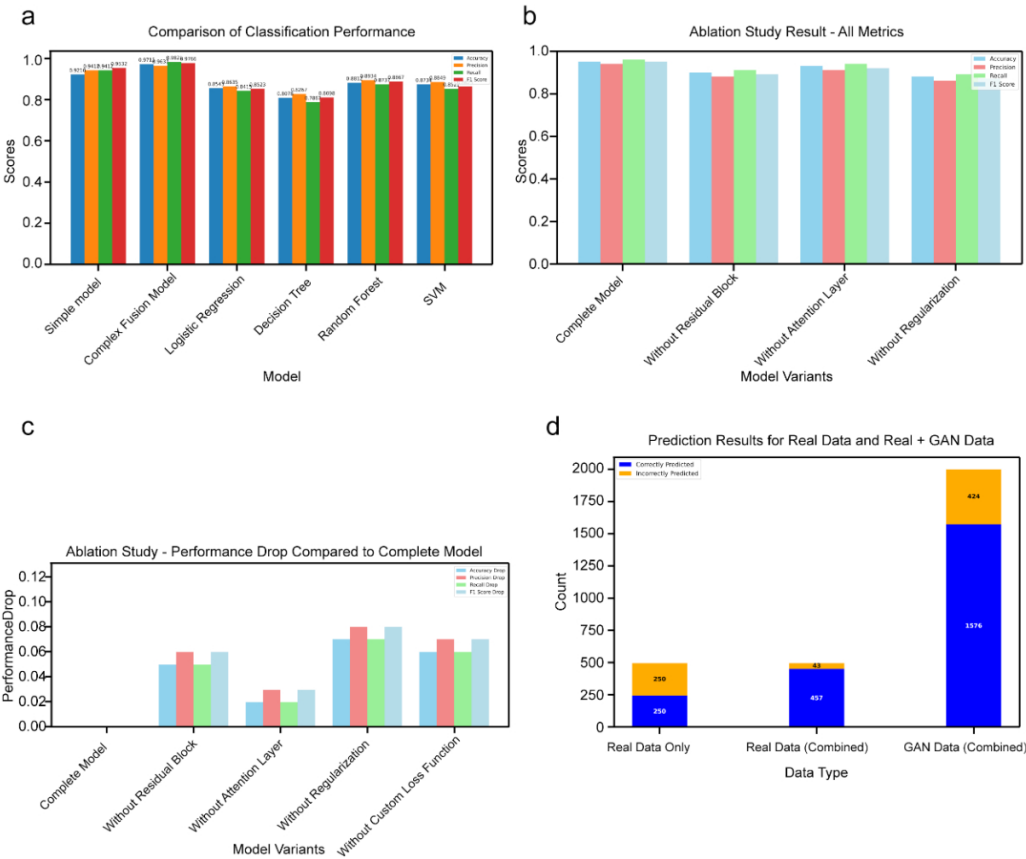


Figure 4: Performance test and comparison, data augmentation using GAN, and ablation experimental result figures(a) Comparing performances of the complex ECCNET and classic models (b, c) Ablation experiment result figures of complex ECCNET (d) Comparison result figure with GAN data augmented model.

for complex genomic sequence analysis, with significant potential for both basic research and precision medicine applications. However, the high heterogeneity and complexity of eccDNA may affect the universality of the prediction. Future research can explore more types of biomarkers to enhance the adaptability and accuracy of models under different biological conditions[1, 18].

Acknowledgments

This work was supported by the Key Research and Development Project of Sichuan Science and Technology Program (24ZDYF0889), Basic Research Cultivation Support Program of Fundamental Research Funds for the Central Universities (2682023ZTPY071) and Sichuan Science and Technology Program under Grant (2022NS-FSC0779).

References

[1] LI R, WANG Y, LI J, *et al.* Extrachromosomal circular DNA (eccDNA): an emerging star in cancer[J]. Biomarker Research, 2022,10(1):53.
[2] ZHU J, ZHANG F, DU M, *et al.* Molecular characterization of cell-free eccDNAs in human plasma[J]. Scientific Reports, 2017,7(1):10968.
[3] CHITWOOD D G, WANG Q, KLAUBERT S R, *et al.* Microevolutionary dynamics of eccDNA in Chinese hamster ovary cells grown in fed-batch cultures under control and lactate-stressed conditions[J]. Scientific Reports, 2023,13(1):1200.
[4] CHANG K-L, CHEN J-H, LIN T-C, *et al.* Short human eccDNAs are predictable from sequences[J]. Briefings in Bioinformatics, 2023,24(3):bbad147.

[5] DEHKORDI S R, WONG I T-L, NI J, *et al.* Breakage fusion bridge cycles drive high oncogene copy number, but not intratumoral genetic heterogeneity or rapid cancer genome change[A]. [2025-05-09].
[6] ZHENG X, ZHAO X, HAN Z, *et al.* ENHANCER OF ZESTE HOMOLOG 2 participates in the process of atherosclerosis by modulating MICRORNA -139-5p methylation and signal transducer and activator of transcription 1 expression[J]. IUBMB Life, 2021,73(1):238–251.
[7] MANN L, SEIBT K M, WEBER B, *et al.* ECCsplorer: a pipeline to detect extrachromosomal circular DNA (eccDNA) from next-generation sequencing data[J]. BMC Bioinformatics, 2022,23(1):40.
[8] PRADA-LUENGO I, KROGH A, MARETTY L, *et al.* Sensitive detection of circular DNAs at single-nucleotide resolution using guided realignment of partially aligned reads[J]. BMC Bioinformatics, 2019,20(1):663.
[9] KUMAR P, DILLON L W, SHIBATA Y, *et al.* Normal and Cancerous Tissues Release Extrachromosomal Circular DNA (eccDNA) into the Circulation[J]. Molecular Cancer Research, 2017,15(9):1197–1205.
[10] WANCHAI V, JENJAROENPUN P, LEANGAPICHART T, *et al.* Accurate Identification of Extrachromosomal Circular DNA from Long-read Sequences[A]. [2025-05-09].
[11] YANG B, BENDER G, LE Q V, *et al.* CondConv: Conditionally Parameterized Convolutions for Efficient Inference[A]. arXiv,2020[2025-05-09].
[12] CHEN Y, DAI X, LIU M, *et al.* Dynamic Convolution: Attention over Convolution Kernels[A]. arXiv,2020[2025-05-09].
[13] PEREZ-LOPEZ R, GHAFARI LALEH N, MAHMOOD F, *et al.* A guide to artificial intelligence for cancer researchers[J]. Nature Reviews Cancer, 2024,24(6):427–441.
[14] ZHONG T, WANG W, LIU H, *et al.* eccDNA Atlas: a comprehensive resource of eccDNA catalog[J]. Briefings in Bioinformatics, 2023,24(2):bbad037.
[15] SUN H, LU X, ZOU L. EccBase: A high-quality database for exploration and characterization of extrachromosomal circular DNAs in cancer[J]. Computational

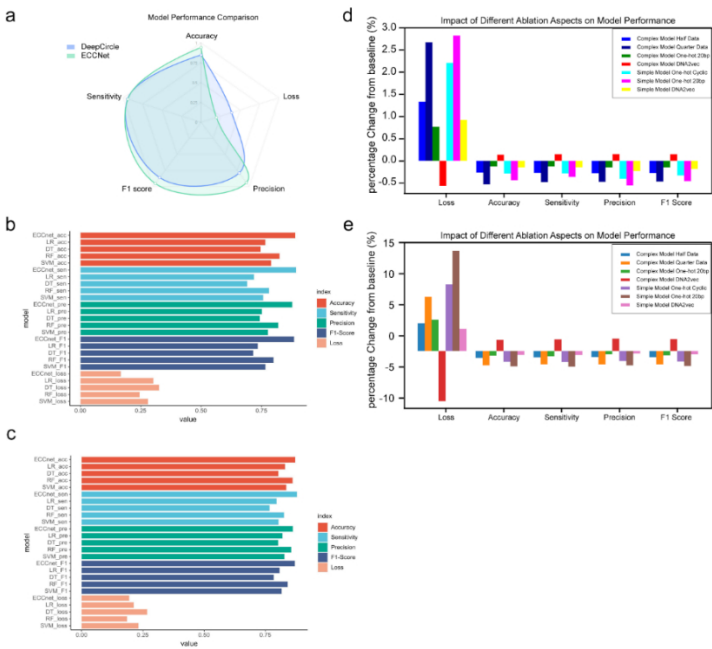


Figure 5: Comparative and ablation experiment analyses. (a) Radar chart comparing ECCNET with similar models (e.g., DeepCircle). (b, c) Bar charts comparing ECCNET with traditional classification models. (d, e) Line plots from ablation experiments.

and Structural Biotechnology Journal, 2023,21:2591–2601.

[16] ZHAO X, SHI L, RUAN S, *et al.*. CircleBase: an integrated resource and analysis platform for human eccDNAs[J]. Nucleic Acids Research, 2022,50(D1):D72–D82.

[17] DALE B, ANDERSON C, PARK K-S, *et al.*. Targeting Triple-Negative Breast Cancer by a Novel Proteolysis Targeting Chimera Degradar of Enhancer of Zeste Homolog 2[J]. ACS Pharmacology & Translational Science, 2022,5(7):491–507.

[18] LUO X, ZHANG L, CUI J, *et al.*. Small extrachromosomal circular DNAs as biomarkers for multi-cancer diagnosis and monitoring[J]. Clinical and Translational Medicine, 2023,13(9):e1393.